

SESSION 9

1. COMMIT

COMMIT makes the insert permanent. After reconnecting, SELECT verifies the order remains.

```
INSERT INTO Orders (order_id,user_id,order_total) VALUES (1001,1,799.00);
COMMIT;
SELECT * FROM Orders WHERE order_id=1001;
```

2. ROLLBACK

ROLLBACK undoes uncommitted inserts. Verify that the rows are absent afterward.

```
START TRANSACTION;
INSERT INTO OrderItems (item_id,order_id,item_name) VALUES (1,1001, 'Pizza'), (2,1001, 'Drink');
ROLLBACK;
SELECT * FROM OrderItems WHERE order_id=1001;
```

3. SAVEPOINT

ROLLBACK TO keeps changes made before the savepoint but removes changes after it.

```
START TRANSACTION;
INSERT INTO Cart (cart_id,user_id,product_name) VALUES (1,1, 'Headphones');
SAVEPOINT after_first_product;
INSERT INTO Cart (cart_id,user_id,product_name) VALUES (2,1, 'Phone Case');
ROLLBACK TO after_first_product;
COMMIT;
SELECT * FROM Cart WHERE user_id=1;
```

4-5. Wallet trigger and test

This MySQL trigger subtracts the transaction amount after a new transaction. The test starts with balance 1000 and records a 250 expense, so the expected balance is 750.

```
DELIMITER //
CREATE TRIGGER deduct_wallet_balance
AFTER INSERT ON Transactions FOR EACH ROW
BEGIN
    UPDATE Wallet SET balance=balance-NEW.amount WHERE user_id=NEW.user_id;
END//
DELIMITER ;
UPDATE Wallet SET balance=1000 WHERE user_id=1;
INSERT INTO Transactions (transaction_id,user_id,amount) VALUES (1,1,250);
SELECT balance FROM Wallet WHERE user_id=1;
```